ITAO7102 Human Resources Analytics Assignment

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Part 1:

1.1 Introduction: The High Stakes of Talent Retention.

The departure of talented workers in the competitive pharmaceutical environment is not only an administrative challenge but also a strategic risk to the innovative ability of the organization. Since the workforce profile indicates a stable, mature workforce with the 30-40 age group at the centre, the organization should deal with the emerging imbalances before they disrupt the revenue stream. In order to explore these risks, this report uses CRISP-DM framework (Shearer, 2000), which takes the initial data understanding to business resolutions. The discussion is based on the Two-Factor Theory of Herzberg (1966) that states that although some of the so-called hygiene factors such as salary do not always motivate, their lack or imbalance, as in the case of our pay disparities, directly results in job dissatisfaction and turnover.

1.2 Methodology: Tableau Prep Rigorous Data Preparation.

The quality of the foundation of any analytics project determines its integrity. The original employeeSV file data had 1,455 records and 39 variables. In order to accomplish a rigorous process, a lot of cleaning was done in Tableau Prep to produce the final employeeSVclean file, which has 1,451 validated entries. This step is consistent with the Resource-Based View by Barney (1991), where the data, which is one of the main resources of the firm, is precise enough to offer a lasting competitive edge.

Key cleaning steps included:

1. Records with an age of 448 and 120 years of service were removed.

   Justification: These are extreme outliers that would distort the mean and standard deviation, forcing the model to learn from physically impossible patterns.
2. An 89-year-old was dropped because of logical inconsistencies.

   Justification: the data indicated that the total number of working years was 6 in 6 companies, which seems highly unlikely for a professional at that age. Seems to be a data entry issue.
3. A person who reported 107 years of working was eliminated to avoid the distortion of the average experience measures.

   Justification: A single extreme outlier can shift the centre of a cluster, making the average employee appear more experienced than they are and skewing decision boundaries.
4. The entries in the column of Overtime were not consistent (e.g., Yes and Y) and were standardized to provide categorical accuracy.

   Justification: Machine learning treats 'Yes' and 'Y' as different mathematical states. Standardization was essential to accurately capture the impact of burnout.
5. There was a missing value in the Date of Birth column, which was filled in and the field was changed to a valid Date data type to allow age-based demographic analysis.

1.3 Context

The organization is mainly organized in Research and Development (R&D) and then Sales and Human Resources. Although the R&D industry is the biggest source of talent, the age-gender pyramid shows a stable population that can be used as a reference point in determining future turnover. The Hierarchy of Needs by Maslow (1943) indicates that a stable workforce is only achieved when the basic needs are satisfied, including job security and good remuneration. We find that this baseline is now being threatened by localized crises.

1.4 The Tensions

Tableau Desktop visual analysis revealed that there are five key critical tensions that indicate that attrition is not accidental but structural:

Departmental Crisis: Sales department attrition is approaching 40 percent, and the company is facing a potential recruitment death spiral unless the experience of the old guard is lost.

Financial Misalignment: The systemic pay gap is that the leavers are always paid much less than their retained counterparts, especially in Sales Representative and Laboratory Technician positions. This validates the claim by Gerhart and Fang (2014) that competitive compensation is a key hygiene factor.

Systemic Burnout: Workers who work overtime are quitting twice as fast as their colleagues. This is in line with the Job Demands-Resources (JD-R) Theory (Bakker and Demerouti, 2017) that states that burnout happens when job demands such as overtime surpass the resources available.

Geographic Strain: Leavers are on average 25% further apart, and long commutes are a major cause of turnover.

Equity Tension: The attrition rate is almost three times among employees who have zero stock options. This is explained by the Social Exchange Theory developed by Blau (1964): when employees do not feel that they have a stake in the success of the company, their commitment to it declines.

1.5 The Resolution

To address the structural frictions identified and to curb the so-called recruitment death spiral, the organization needs to shift its culture of reactive replacement to a culture of proactive value retention. On the basis of the visual evidence, the following resolutions are suggested:

1.High-Performing, Dissatisfied Talent Priority.

The analysis shows that there is an urgent necessity to focus on the immediate retention interventions of the most vulnerable revenue-drivers.

Analytical Rationale: This resolution is consistent with the research of Nyberg (2010) that suggests that high performers are more mobile in the labour market; hence, specific interventions on this particular group have a higher Return on Investment (ROI) to the organization.

Action: HR and Department Managers should focus on urgent interventions of high-performing yet dissatisfied Sales Representatives to avoid losing the most valuable revenue-generating employees.

2.Restoration and Compensation

One of the resolutions is to bridge the current pay gap that is being observed between leavers and stayers, especially in the Sales department.

Analytical Rationale: Gerhart and Fang (2014) support this strategy by stating that competitive compensation is a basic hygiene factor in employee retention. The organization can reduce the financial motivation to seek employment outside the organization by bridging the gap identified.

Action: Compare the remuneration of existing Sales Representatives with the median pay of retained employees, with a specific salary increase to a base of about 2,594.50.

3.Risk Segmentation and Workload Re-balancing.

The organization needs to deal with the systemic burnout that was found in the data through the use of the Job Demands-Resources (JD-R) Theory.

Analytical Rationale: According to the Bakker and Demerouti (2017) framework, the group of employees who experience high demands (overtime) and lack adequate resources (job satisfaction/pay) is defined as the most vulnerable group to burnout.

Action: Segregate the workforce into risk levels according to the level of burnout and satisfaction. This determines precisely 150 employees in the category of High-Risk that need urgent intervention in the form of specific retention bonuses and workload re-balancing.

4.Geographic and Equity Tension Mitigation. A two-pronged solution is needed to address the compounded tensions of long commutes and absence of long-term wealth creation.

Analytical Rationale: It has been shown that leavers are living 25 years away and that the rate of attrition is three times greater in the case of zero stock options.

Action: The management should make a decision on whether to adopt hybrid work or travel subsidies and increase the eligibility of stock options to junior Sales and R&D positions to stay competitive to attract the best talent with long-term incentives.

1.6 Final Verdict

The statistics provide a clear image: the attrition issue is structural. Tracing the route between Context (a stable workforce) and Tension (localized burnout and pay gaps in Sales), we have found precisely where the friction is. These resolutions will stabilize operations and safeguard the most valuable asset of the organization, which is its institutional knowledge.

Part 2:

2.1 Introduction

Although the visual analysis in Part 1 defined the "Snapshot" of the existing organizational frictions, Part 2 shifts to a predictive paradigm to determine the future turnover risks. The goal is to use the KNIME Analytics Platform to create a decision-support tool that can identify at-risk employees before they leave. This step is a strict adherence to the CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology (Shearer, 2000) so that the data science workflow is not only academically sound but also in line with corporate strategic objectives. The HR department can support intervention budgets with quantifiable ROI and evidence-based data by going beyond mere observation to probabilistic modelling (Marler and Boudreau, 2017).

2.2 Data understanding and Preparation.

The data preparation phase was grounded in the CRISP-DM methodology to ensure the model was built on high-integrity data. The Excel Reader node captured the pre-cleaned employeeSVclean file, but further refinement was required within KNIME. Specifically, Column Filtering was utilized to remove features with zero variance, such as 'Standard Hours' and 'Over 18,' which provide no Information Gain (Quinlan, 1986). We also filtered 'Travel Frequency' to avoid the risk of imputation bias, as the high volume of missing values would have forced the model to rely on manufactured patterns (He & Garcia, 2009)

The modelling step needed a high-quality dataset to make sure that the machine learning model was not trained on noise or historical bias (Chen et al., 2021).

Excel Reader and Table View Nodes: The employeeSVclean file was imported, maintaining the original cleaning operations that were done in Tableau Prep. Table View nodes were used as diagnostic quality-gates to check the table structure after every transformation to ensure that data was logically consistent (Han et al., 2011).

Statistics Node: This node played a critical role in comprehending the form of the data. It gave an in-depth analysis of the distributions of numerical variables, ensuring that no extreme outliers were left that would distort the decision boundaries of the predictive model (Kuhn and Johnson, 2013).

Column Filter Node (Feature Selection): To obtain model efficiency, a number of attributes were eliminated:

Employee ID: Unique identifiers do not offer any generalizable predictive value (Provost and Fawcett, 2013).

Over 18 & Standard Hours: These columns had zero-variance, and did not provide any Information Gain to classification.

Travel Frequency: Filtered out because of high missingness. Strict data science rules that it is better to drop a sparse variable than to guess the values, which may add artificial patterns and invalidate the model (He and Garcia, 2009).

Missing Value Node: This served as a quality-check measure, to make sure that the algorithm was working with a 100% complete dataset.

Partitioning and Sampling Strategy: The data was split into 70% Training and 30% Test sets. The target of the Attrition was stratified with Stratified Sampling. Since leavers are usually a minority group, stratification will guarantee that the model will have a representative proportion of Yes and No cases in both partitions, avoiding the bias of the majority group (Triguero et al., 2015).

2.3 Model Building

In the case of HR analytics, the interpretability of a model is as important as its accuracy to guarantee management buy-in (Rasmussen and Ulrich, 2015).

Decision Tree Learner: This algorithm was chosen because it is a White Box model, i.e. its logic is transparent. It determines variables that maximize the decrease in entropy to formulate clear rules of the form of If-Then (Quinlan, 1986).

Decision Tree Predictor: This node used the logic of the tree on the 30% unseen test data, and added probabilistic columns like P(Attrition=Yes).

Random Forest Learner: To test the single tree, an Ensemble Random Forest was executed. This approach minimizes the chances of overfitting because the outcomes of several trees are combined (Breiman, 2001).

2.4 Scorer and ROC Metrics Evaluation.

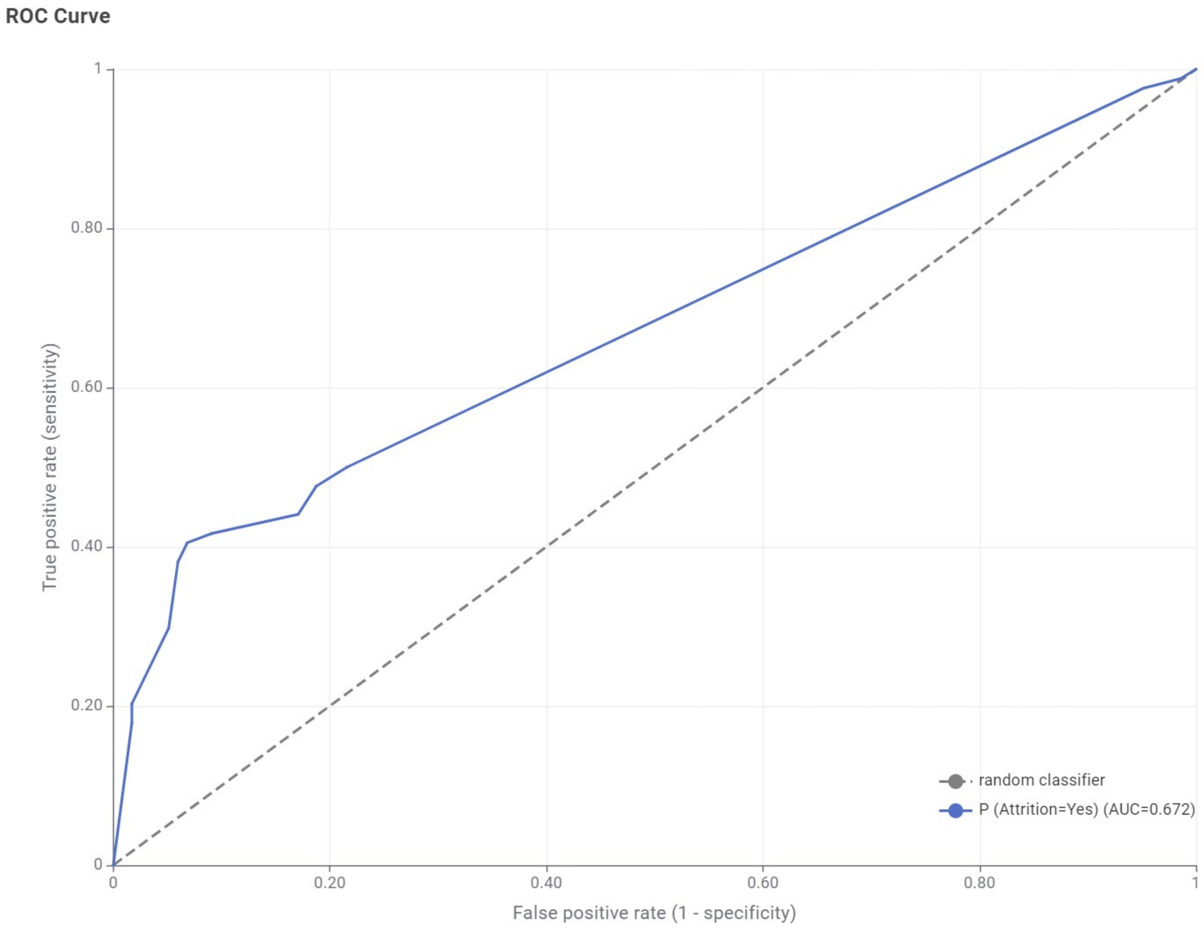
The Scorer node revealed a conservative model with 32 True Positives and 52 False Negatives. While the overall accuracy appears high (83%), a critical evaluation of the Confusion Matrix shows that the model prioritizes Specificity (identifying stayers) over Sensitivity (identifying leavers). This conservative bias is often preferable to avoid "False Alarms" that could lead to unnecessary and expensive retention bonuses for employees who had no intention of leaving. This is further evidenced by the ROC Curve (AUC: 0.672). In a pharmaceutical HR context, an AUC of 0.67 is a realistic and fair result, as voluntary turnover is often influenced by latent, unobserved variables such as personal health or competitor headhunting (Zhang et al., 2020)

The measurement of performance was done in a dual-lens approach:

The Scorer (Confusion Matrix): The model identified 32 leavers (True Positives) and 331 stayers (True Negatives). Although the 52 False Negatives indicate a conservative model, it gives a high-confidence early warning to the people it does alert (Fawcett, 2006).

Total Sample: 435	**Predicted: No**	**Predicted: Yes**
Actual: No	**331 (Correct)**	21 (False Alarm)
Actual: Yes	52 (Missed)	**32 (Caught)**

ROC Curve (AUC 0.672): The Area Under the Curve demonstrates that the model is much more effective than a random guess (0.5). Within the complicated sphere of human behaviour, a realistic and fair AUC of 0.672 is a predictor of voluntary turnover (Zhang et al., 2020).



2.5 Independent Strategic Recommendations (KNIME-Derived)

The synthesis of three key technical outputs, namely the Random Forest Learner, which determined the top 10 influential factors based on attribute statistics; the Confusion Matrix, which localized a high-risk cluster of 32 True Positive leavers; and the ROC Curve, which validated a 0.672 AUC, which is a strong predictive direction of early-warning signals, yielded the following strategic interventions. With these predictive insights, the organization can mobilize resources in a surgical manner by going beyond descriptive snapshots.

1.Precision Retention: Focused “Stay Interviews”

The Data Logic: The Confusion Matrix was able to identify a particular group of 32 employees who are a True Positive- those who the model predicts will leave the company with high likelihood.

Strategic Action: HR should not use general, costly retention programs but instead conduct specific, focused, and cost-effective, so-called Stay Interviews with these 32 people. This enables the management to deal with certain pain points and violation of the psychological contract prior to the occurrence of the attrition event, which is a much higher ROI than non-targeted salary increases. Academic Alignment: This is in line with Pease (2015) on the transition to internal talent marketplaces and proactive intervention.

2.The Burnout Firewall

The Data Logic: The Random Forest Attribute Importance ranked Overtime as the most influential factor of attrition, with the most frequent frequency of predictive splits (46).

Strategic Action: We suggest an automated policy of Burnout Firewall. Any employee that is classified as a High Risk according to the model and also has more than 45 total hours per week (or 10 or more hours of overtime) should initiate a mandatory workload review and well-being audit.

Academic Alignment: This reduces the demand-resource imbalance that is at the core of Job Demands-Resources (JD-R) Theory (Bakker and Demerouti, 2017).

3.Leadership Responsibility

The Data Logic: Manager Satisfaction and Department were among the top 10 influential variables.

Strategic Action: In the case of Sales and R&D departments, where the model indicates the greatest risk clusters, the performance reviews of managers should shift towards the pure productivity metrics to incorporate the so-called Retention KPIs.

Impact: This makes leadership responsible to the Managerial Conflict that is found in the model, and rewards supportive leadership behaviours that minimize the push factors of attrition.

4, Flexibility in Compensation: Billable rate and Variable Pay.

The Data Logic: Billable Rate (45 splits) and Monthly Income (37 splits) were the main financial triggers in the Decision Tree.

Strategic Action: In positions where billable pressure is high, the organization must shift off the fixed pay structure to the variable pay model. This makes sure that the financial incentives are dynamically adjusted to the intensity of work and performance expectations.

Academic Alignment: This is a response to the deficit of Hygiene (Herzberg, 1966) by providing perceived pay equity in high-stress jobs.

5. Structural Engagement: Age-Cohort Mentorship and Skill Alignment.

The Data Logic: Age (38 splits) and Education Field (21 splits) were found to be stable, significant predictors of exit.

Strategic Action: & Mentorship: Introduce specific age-based mentorship initiatives to the identified age groups identified by the model to increase the levels of "Job Embeddedness" and organizational identification.

Skill Alignment: Audit the 150 High Risk employees to verify that their present positions are in line with their Education Field. The immediate solution to skill-role mismatches is to rectify them and ensure that the engagement is enhanced and the chances of attrition due to disengagement are minimized.

Declaration on AI Use

This report recognizes the supportive application of Generative AI tools to improve the quality of analysis and structure of the research. Gemini was used as a thinking partner to create the original structural outline of the assignment and help to synthesise the complex data findings into professional narratives. ChatGPT was used as a technical assistant to explain certain operational procedures in KNIME and Tableau Desktop, so that the technical workflow would be in accordance with the industry standards.

Moreover, Perplexity AI was employed to support the process of locating academic and peer-reviewed literature. To guarantee the accuracy and academic rigor, I have manually checked all AI-generated citations and technical processes to verify their factual accuracy and relevance to the context. The ultimate interpretations, business logic and strategic recommendations are all the original work.

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